Pixels, Frames & Transformers

Images as Data · Module 01

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Why Should Social Scientists Care About Images?

A growing share of political, health and science communication is visual. Some examples:

Protest and mobilization
which protest photos spread and mobilize; detecting protest in social media images (Casas & Williams, 2019; Won et al., 2017)

Health and food marketing
health campaigns, packaging aimed at children, influencers' nutrition advice

Political self-presentation
how news photos portray candidates; how politicians show themselves on Instagram (Peng, 2018, 2021)

Misinformation and news
manipulated or out-of-context images, and whether people believe them (Hameleers et al., 2020)

This course goes from pixels to models that see, read and listen. Module 01 starts with what a computer actually sees.

An Image Is a Grid of Numbers



228 185 176	227 183 174	225 181 168	222 178 165
227 184 175	227 183 172	224 180 167	223 177 164
223 179 170	222 178 165	222 176 163	220 172 158
216 168 158	215 167 153	216 164 150	214 163 146

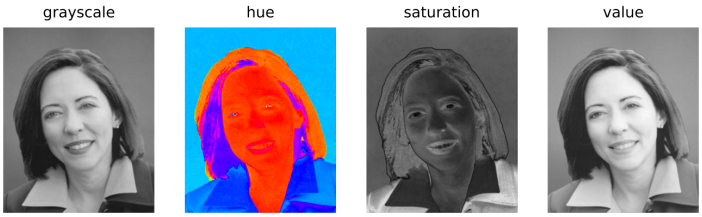
4 × 4 pixels at the corner of the mouth, each with its red, green and blue value

- A color photo is three tables of numbers, the **channels**: red, green, blue
- Values run from **0** (no light) to **255** (full intensity), as `uint8`
- In NumPy this portrait has shape `(550, 450, 3)`: height, width, channels

A classic bug: Pillow gives sizes as (width, height), NumPy as (height, width).

Official portrait of Sen. Maria Cantwell, public domain, via the unitedstates/images project.

Color Spaces: Choose What Is Easy to Measure



RGB
how screens make color, and the format most files use

Grayscale
a weighted average:
$L = 0.299R + 0.587G + 0.114B$

HSV
which color (hue), how vivid (saturation), how bright (value)

How vivid is an image, whatever its colors? In HSV that is a single channel.

Editing Is Arithmetic (and `uint8` Bites)

original



arr + 100



clipped to 0-255



Brighten by adding 100?

```
arr + 100              # 235 + 100 = 79 !
np.clip(arr.astype(int) + 100,
        0, 255).astype(np.uint8)
```

`uint8` values silently **wrap around** above 255. Compute with a larger type, clip, convert back.

- Cropping and flipping are just indexing
- Resizing computes new pixels: shrinking loses detail, enlarging can't bring it back

Same File Size, Different Detail

230 × 281, enlarged



675 × 825, reduced



Two files from the same project, both 450 × 550. The left one was enlarged from a 230 × 281 original.

Saving the 450 × 550 portrait

format	file size	pixel change
PNG (lossless)	268 KB	0.0
JPEG quality 95	52 KB	0.9
JPEG quality 50	14 KB	2.1
JPEG quality 5	6 KB	9.4

A uniform-looking collection can hide large differences in resolution and compression, and web images are recompressed at many uploads. **Find out how your images were produced before you compare them.**

Official portraits of Sen. Maria Cantwell and Sen. Amy Klobuchar, public domain, via unitedstates/images.

Metadata: What Else Is in the File

EXIF in the downloaded “original” file

```
Software      Adobe Photoshop CS3 Macintosh
DateTime      2009:02:06 13:22:10
ImageWidth    675
ImageLength   825
```

. . . but the file itself is 230 × 281. Metadata often describes an earlier version of the file: a useful clue, a risky fact.

A phone photo can reveal
the phone model, the exact time, the **GPS location** where it was taken, and more. Treat it as personal data and strip it before you share anything.

The orientation tag
Phones often store pixels sideways plus a tag saying how to rotate them. Viewers obey it; many pipelines don’t, and quietly analyze a sideways image. Apply `ImageOps.exif_transpose()` first.

From One Image to a Collection



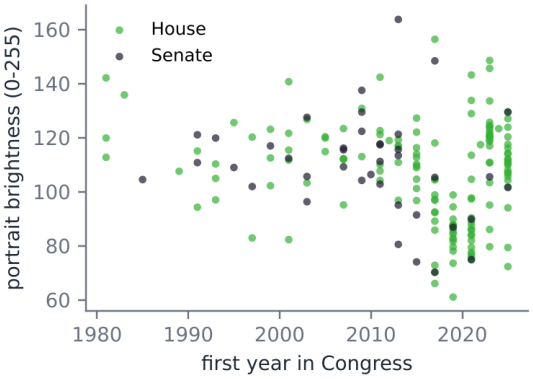


The average of 193 portraits, pixel by pixel

A **contact sheet** is the quickest way to look at a collection, and you should always look before you measure. **Averaging** shows a genre's conventions: what every image shares (pose, framing, dark suit, blue backdrop) stays sharp; what varies blurs away.

Official portraits of current members of the US Congress, public domain, via unitedstates/images. Random sample of 200, of which 193 had a portrait.

A Measure That Tracks the Photos, Not the People



Average brightness of each portrait

- Members who entered Congress in **2017–2021**: 92 on average
- Every other period: 112 to 114
- Democrats 106, Republicans 108

Brightness here mostly measures *when and how the photo was taken*: the studio, the backdrop, the photographer, the year. A party gap could arise from production alone.

Before interpreting a visual measure, ask what process produced the images, and compare like with like.

Content, Form and Meaning

Content
What is in it?
people, objects, text, setting

Computers have become very good at this.

Form
How does it look?
color, light, composition, framing

Easy to compute, often hard to interpret.

Meaning
What does it say?
frames, emotion, symbolism, persuasion

Needs a codebook, human judgment and validation.

Good computational image research uses the computer for content and form at a scale no team of coders could manage, and links those measures to meaning through a codebook and validation, just as in text analysis.

In the Notebook

- Pixels, channels and color spaces: an official portrait as a NumPy array
- Editing by arithmetic: the `uint8` trap, crops, flips and resizing
- Resolution and JPEG quality: what files don't tell you
- Metadata: reading EXIF, a fake phone photo with GPS, removing it safely
- 200 portraits: a contact sheet and the average face of Congress
- A first measurement, and what it really measures

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